

Practice 1: Metric length

Name: _____ Date: _____

Use the metric relationships. Show your work and include units.

1. Convert 6 meters to centimeters.

Work space

2. Convert 4,300 millimeters to meters.

Work space

Practice 2: Metric mass

Name: _____ Date: _____

Remember: 1 kilogram = 1,000 grams and 1 gram = 1,000 milligrams.

1. Convert 7 kilograms to grams.

Work space

2. Convert 5,600 grams to kilograms.

Work space

Practice 3: Metric capacity

Name: _____ Date: _____

Use 1 liter = 1,000 milliliters.

1. Convert 9 liters to milliliters.

Work space

2. Convert 2,750 milliliters to liters and milliliters.

Work space

Practice 4: Customary length

Name: _____ Date: _____

Use $12 \text{ inches} = 1 \text{ foot}$, $3 \text{ feet} = 1 \text{ yard}$, and $5,280 \text{ feet} = 1 \text{ mile}$.

1. Convert 7 feet to inches.

Work space

2. Convert 5 yards to feet.

Work space

Practice 5: Customary mass and capacity

Name: _____ Date: _____

Use 16 ounces = 1 pound, 2 cups = 1 pint, 2 pints = 1 quart, and 4 quarts = 1 gallon.

1. Convert 6 pounds to ounces.

Work space

2. Convert 3 gallons to quarts.

Work space

Practice 6: Converting mixed units

Name: _____ Date: _____

Rewrite each measurement in the requested single unit.

1. Convert 4 feet 7 inches to inches.

Work space

2. Convert 3 yards 2 feet to feet.

Work space

Practice 7: Decimal conversions

Name: _____ Date: _____

Use place value to convert. Write a decimal when it makes sense.

1. Convert 3.6 meters to centimeters.

Work space

2. Convert 475 centimeters to meters.

Work space

Practice 8: Choosing a useful unit

Name: _____ Date: _____

Choose the most reasonable unit, then explain briefly.

1. Would you measure the length of a pencil in inches, feet, or miles? Explain.

Work space

2. Would you measure the amount of water in a bathtub in milliliters, liters, or kiloliters? Explain.

Work space

Practice 9: Comparing measurements

Name: _____ Date: _____

Convert to the same unit before comparing. Use $>$, $<$, or $=$.

1. 3 meters ____ 280 centimeters

Work space

2. 2 kilograms ____ 1,950 grams

Work space

Practice 10: Adding and subtracting measurements

Name: _____ Date: _____

Line up units or convert first.

1. Find $2\text{ m } 45\text{ cm} + 1\text{ m } 80\text{ cm}$. Give the answer in meters and centimeters.

Work space

2. Find $5\text{ kg } 200\text{ g} - 2\text{ kg } 750\text{ g}$. Give the answer in grams.

Work space

Practice 11: Multiplying and dividing measurements

Name: _____ Date: _____

Solve and label each answer.

1. Four boards are each 1.25 meters long. What is their total length in meters?

Work space

2. A 6-liter jug is shared equally among 8 bottles. How many liters go in each bottle?

Work space

Practice 12: Time-unit conversions

Name: _____ Date: _____

Use $60 \text{ seconds} = 1 \text{ minute}$, $60 \text{ minutes} = 1 \text{ hour}$, and $24 \text{ hours} = 1 \text{ day}$.

1. Convert 7 minutes to seconds.

Work space

2. Convert 3 hours 25 minutes to minutes.

Work space

Practice 13: Elapsed time

Name: _____ Date: _____

Find how much time passes. Show a timeline or count by hours and minutes.

1. A movie starts at 6:35 p.m. and ends at 8:20 p.m. How long is it?

Work space

2. Practice begins at 3:48 p.m. and lasts 1 hour 27 minutes. When does it end?

Work space

Practice 14: Perimeter and measurement

Name: _____ Date: _____

Find the perimeter. Remember to use the same unit on every side.

1. A rectangle is 8 cm long and 5 cm wide. Find its perimeter.

Work space

2. A garden has sides 4 m, 7 m, 4 m, and 7 m. Find its perimeter.

Work space

Practice 15: Area and square units

Name: _____ Date: _____

Area measures the space inside a rectangle. Use $A = \text{length} \times \text{width}$.

1. Find the area of a rectangle 9 cm by 4 cm.

Work space

2. A floor is 12 ft long and 8 ft wide. How many square feet does it cover?

Work space

Practice 16: Volume and cubic units

Name: _____ Date: _____

For a rectangular prism, use $V = \text{length} \times \text{width} \times \text{height}$.

1. Find the volume of a box 5 cm by 3 cm by 2 cm.

Work space

2. A fish tank is 10 in long, 4 in wide, and 6 in high. Find its volume.

Work space

Practice 17: Temperature

Name: _____ Date: _____

Read the scales carefully. For these problems, use whole-number temperatures.

1. The temperature is 18°C in the morning and 25°C in the afternoon. How many degrees did it rise?

Work space

2. The freezer is -12°C . It warms by 7°C . What is the new temperature?

Work space

Practice 18: Multi-step measurement problems

Name: _____ Date: _____

Plan the unit conversions before calculating.

1. A runner completes 3 laps of a 400-meter track and then runs 250 meters. How many kilometers does the runner travel?

Work space

2. A baker has 5 kg of flour and uses 750 g. How many grams remain?

Work space

Practice 19: Measurement data

Name: _____ Date: _____

Use the data to answer each question. Show calculations.

1. Three packages weigh 1.2 kg, 850 g, and 2 kg. What is their total mass in grams?

Work space

2. Four students measure a table: 120 cm, 1.35 m, 118 cm, and 1.22 m. Which measurement is longest? Give it in centimeters.

Work space

Practice 20: Mixed review

Name: _____ Date: _____

Solve each problem. Check that your final unit answers the question.

1. Convert 4.75 meters to centimeters.

Work space

2. A hike begins at 8:40 a.m. and ends at 11:15 a.m. How long is the hike?

Work space